

Vaidehi Gohil

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EDUCATION

Northeastern University, Boston, MA

Expected May 2027

Master of Science in Electrical and Computer Engineering (Concentration: Computer Systems and Software)

Relevant Courses: Wireless Sensors, Fundamentals of Computer Engineering, High Performance Computing, Deep Learning Embedded Systems

Vidyavardhi's College of Engineering and Technology, Mumbai, India

June 2024

Bachelor of Technology in Electronics and Telecommunication Engineering

Relevant Courses: Microcontrollers, Optical Communication Systems, Artificial Neural Networks, Linear Integrated Circuits, Digital Signal Design, DVLSI

TECHNICAL SKILLS

Embedded & Edge: Jetson Orin Nano, Raspberry Pi, ESP-WROOM, Arduino, Pi Pico, MSP430

Languages/OS: C/C++, Python, Matlab, Linux (shell, systemd/services), Windows

Interfaces: I²C, SPI, UART, RS-485, BLE, ROS 2

Analog/EDA: MOSFET and ICs basics, Power-Path switching (BMS), Eagle(studied)

WORK EXPERIENCE

Faclon Labs, Mumbai, India

November 2024 – August 2025

IoT Engineer

- Built a fault-tolerant telemetry service on OpenWrt with Python Modbus (RTU/TCP) poller and TCP services with timeout validation, retry/backoff, and watchdog auto-recovery; deployed across 5+ remote sites to keep data collection running with minimal manual intervention
- Integrated heterogeneous devices (I²C, SPI, UART, RS-485, BLE), defined register maps and polling schedules, and standardized outputs into structured JSON published via MQTT and HTTP for consistent dashboards and faster field diagnostics
- Validated in-house BMS PCB power-path switching between battery and external supply, which was done using three MOSFET switching stage, ensuring expected state transitions and basic functionality during system bring-up

PROJECT

Warehouse Drone Autonomy Stack, Personal Project

November 2025 – Present

- Building an indoor autonomy system for warehouse inventory in GPS-denied navigation now, with a path to barcode-scanning workflows later
- Developing the navigation pipeline on Jetson with ROS 2 and PX4 (MAVLink), which will include sensor ingestion, state estimation, and odometry integration into mapping for localization and navigation
- Hardening system reliability by debugging timing and synchronization, validating TF frame consistency, and tuning ROS 2 QoS across the pipeline to make behavior repeatable across runs

Telehealth, Vidyavardhini's College of Engineering and Technology

August 2023 – January 2024

- Designed and prototyped a telehealth kiosk for rural settings, integrating ECG, SpO₂, and temperature sensing for basic checkups and remote consultation
- Implemented a two-tier architecture with ESP-WROOM for sensor calibration, and has a Raspberry Pi with a web dashboard for real-time visualization and remote video consults
- Wrote sensor drivers and data aggregation code to produce stable readings and reliable delivery to the dashboard, documented system design with calibration and usage procedures for handoff and future deployment

Leadership and Activities

Texas Instruments Innovation Lab, Vidyavardhini's College of Engineering and Technology

August 2023 – May 2024

Secretary

- Coordinated lab operations and workshop logistics for Grade 10–12 students from multiple Mumbai schools, including scheduling, materials setup, and maintaining clear student-facing documentation
- Supported hands-on sessions using TI Tiva C and MSP boards and RSLK robot assembly/bring-up, helping teams troubleshoot basic hardware/firmware issues during activities